

Physics of Semiconductor Devices

ECE210: PSD



INDRAPRASTHA INSTITUTE *of*
INFORMATION TECHNOLOGY
DELHI

Dr. Ram Krishna Ghosh

(week13)



MOS Capacitor and MOSFET



MOSFET

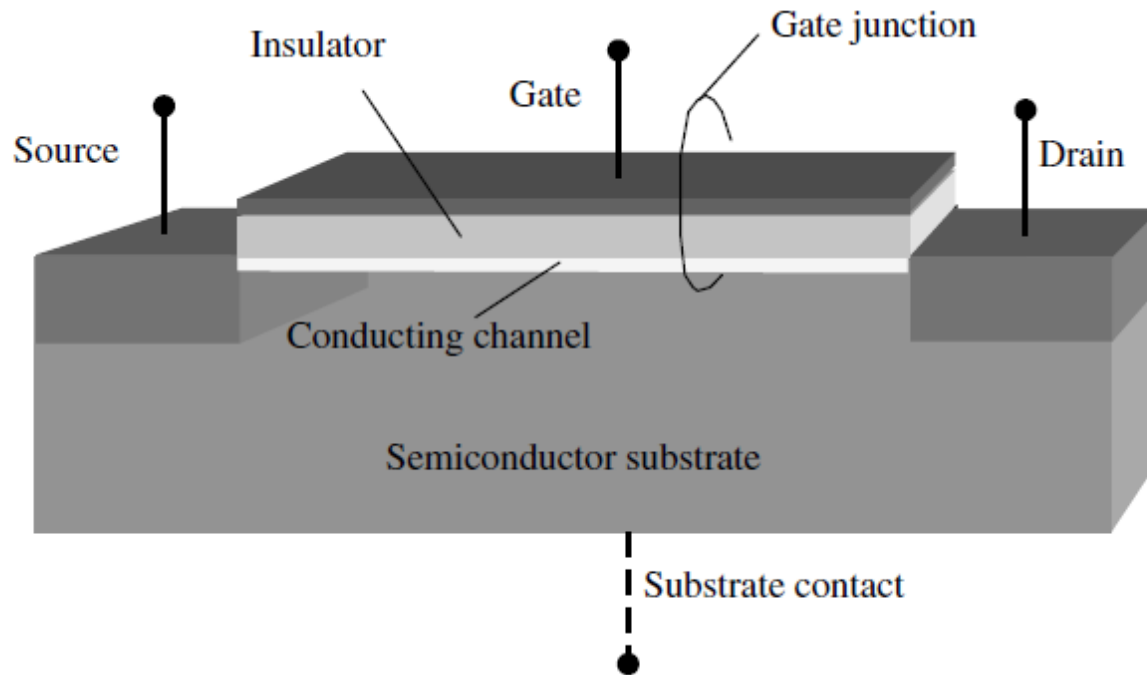
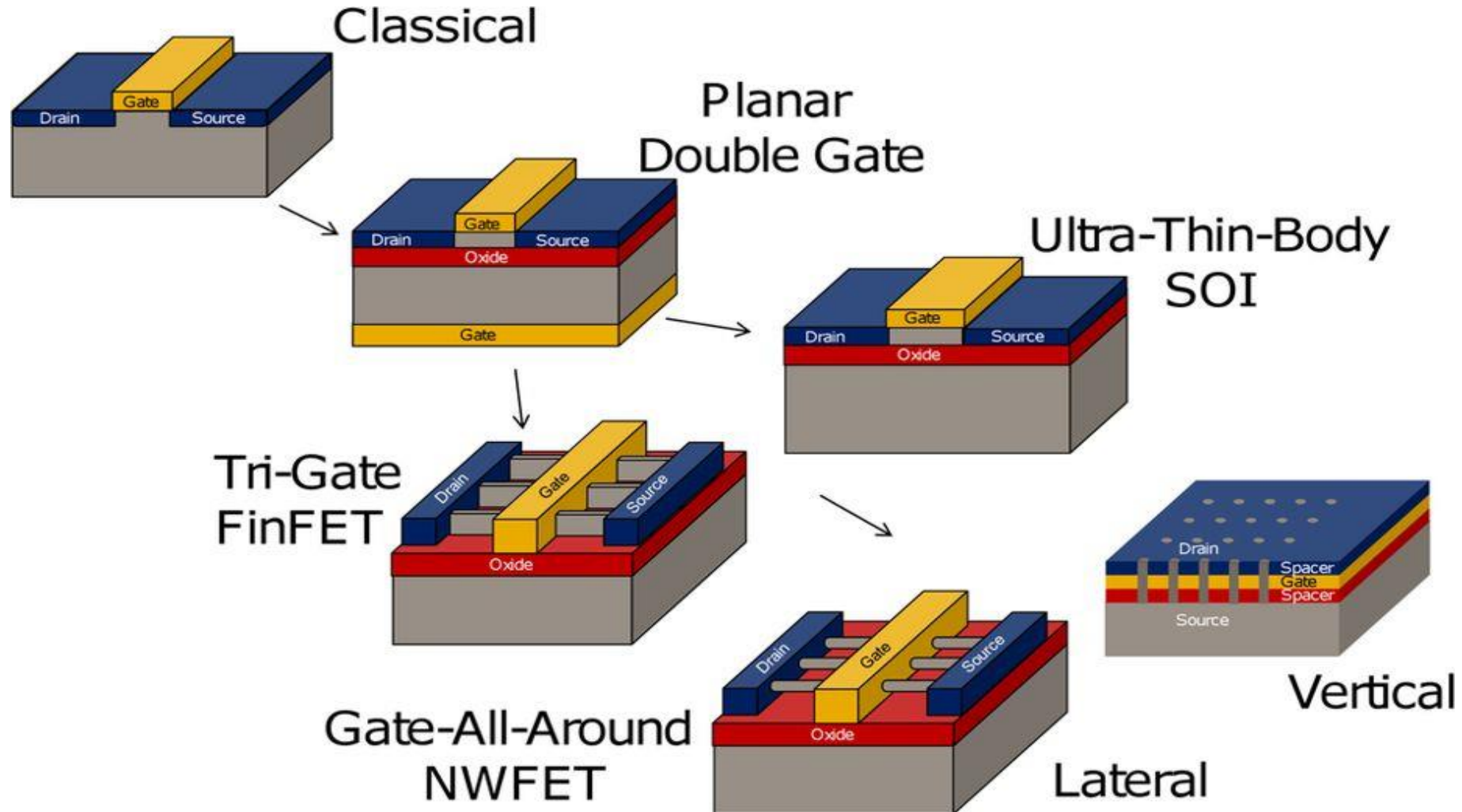


Figure 1.1 Schematic illustration of a generic field effect transistor. This device can be viewed as a combination of two orthogonal two-terminal devices

The charge carriers of the conducting channel constitute an inversion charge, that is, electrons in the case of a p -type substrate (n -channel device) or holes in the case of an n -type substrate (p -channel device), induced in the semiconductor at the silicon-insulator interface by the voltage applied to the gate electrode.

The operation in most regimes is controlled by the channel in MOSFETs – The channel of a MOSFET is an example of a **MOS capacitor**

Different MOSFET architectures: technology trend



MOSCAP



What is the structure of a MOS capacitor?

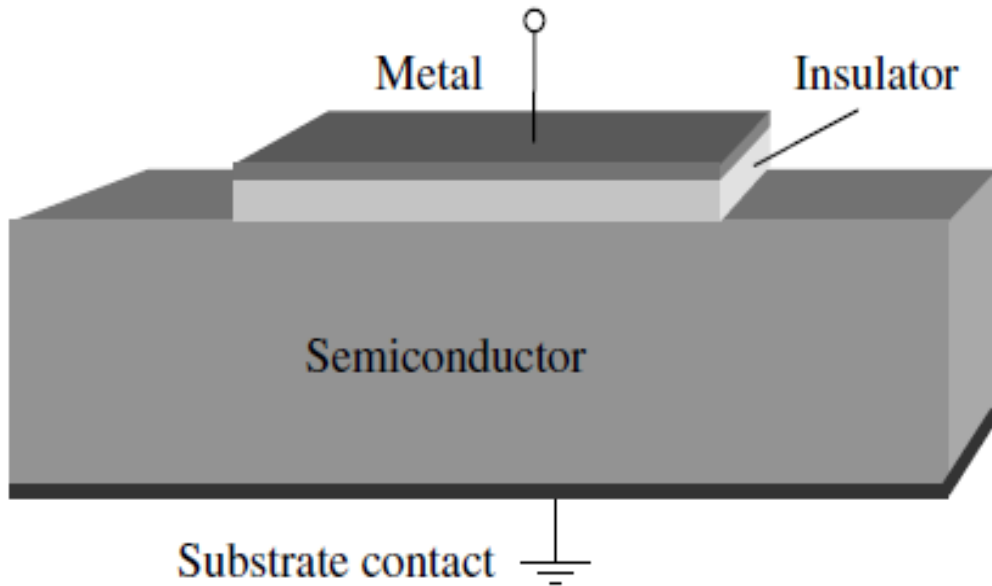


Figure 1.2 Schematic view of a MOS capacitor

Heavily doped polycrystalline Si film as the gate electrode material.

- N-type for “n-channel” transistors (NMOS).
- P-type for “p-channel” transistors (PMOS).

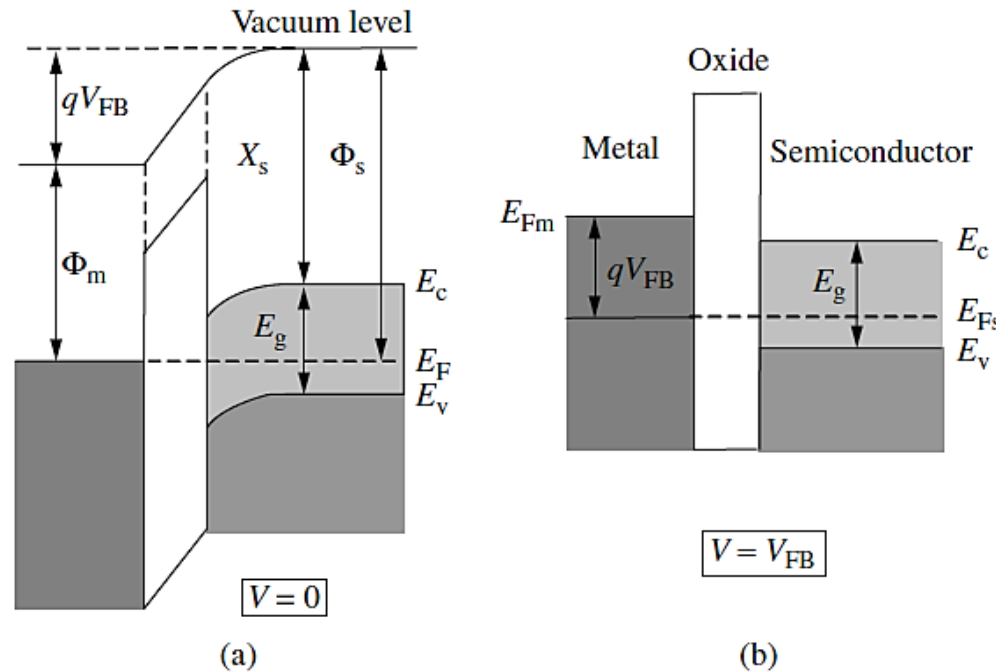
SiO₂ as the gate dielectric

- Band gap = 9 eV.
- Relative dielectric constant $\epsilon_r = 3.9$

Si as the semiconductor material.

- P-type for n-channel devices.
- N-type for p-channel devices.

MOSCAP



At $V=0$, the bending of the energy bands is ideally determined by the difference in the work functions of the metal and the semiconductor.

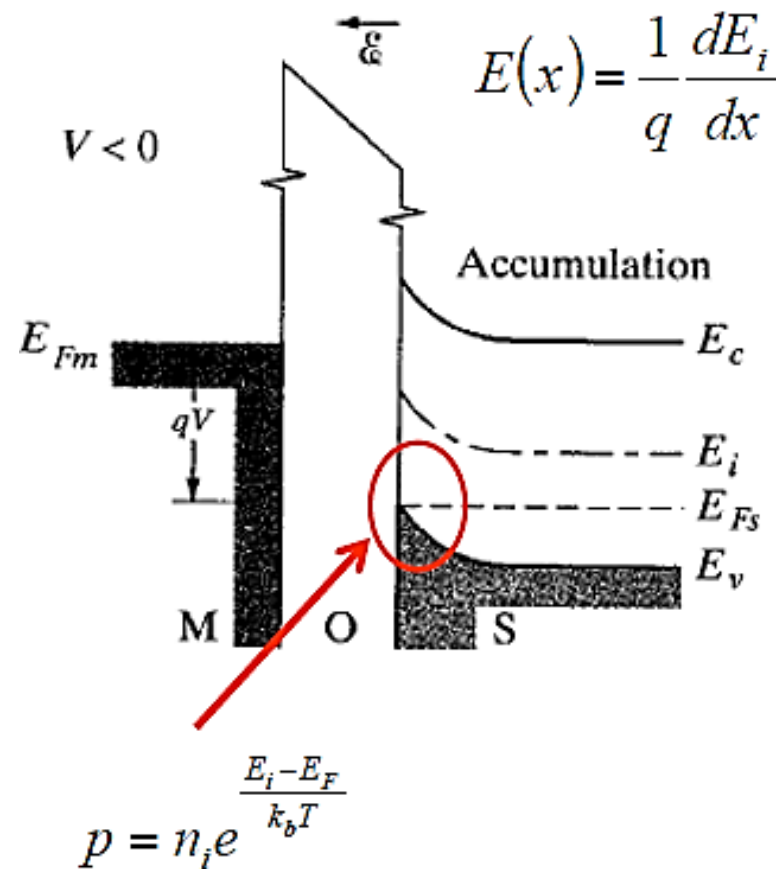
$$V_{FB} = (\Phi_m - \Phi_s)/q = (\Phi_m - X_s - E_c + E_F)/q,$$

In real devices, the flat-band voltage may be affected by surface states at the semiconductor–oxide interface and by fixed charges in the insulator layer

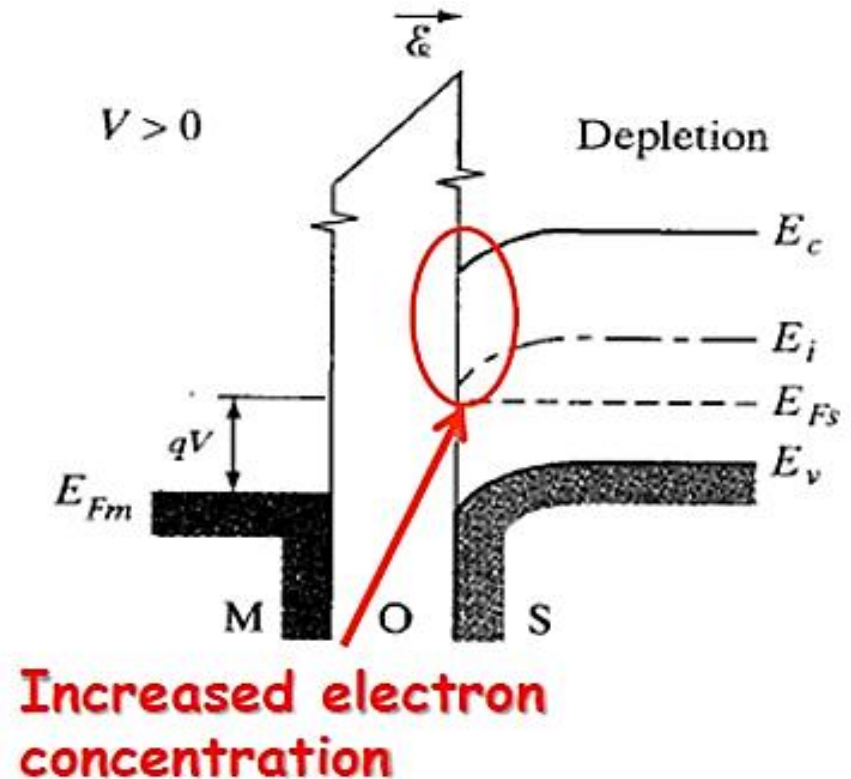
Figure 1.3 Band diagrams of MOS capacitor (a) at zero bias and (b) with an applied voltage equal to the flat-band voltage. The flat-band voltage is negative in this example

At stationary conditions, no net current flows in the direction perpendicular to the interface owing to the very high resistance of the insulator layer. Hence, the Fermi level will remain constant inside the semiconductor, independent of the biasing conditions.

MOSCAP under bias

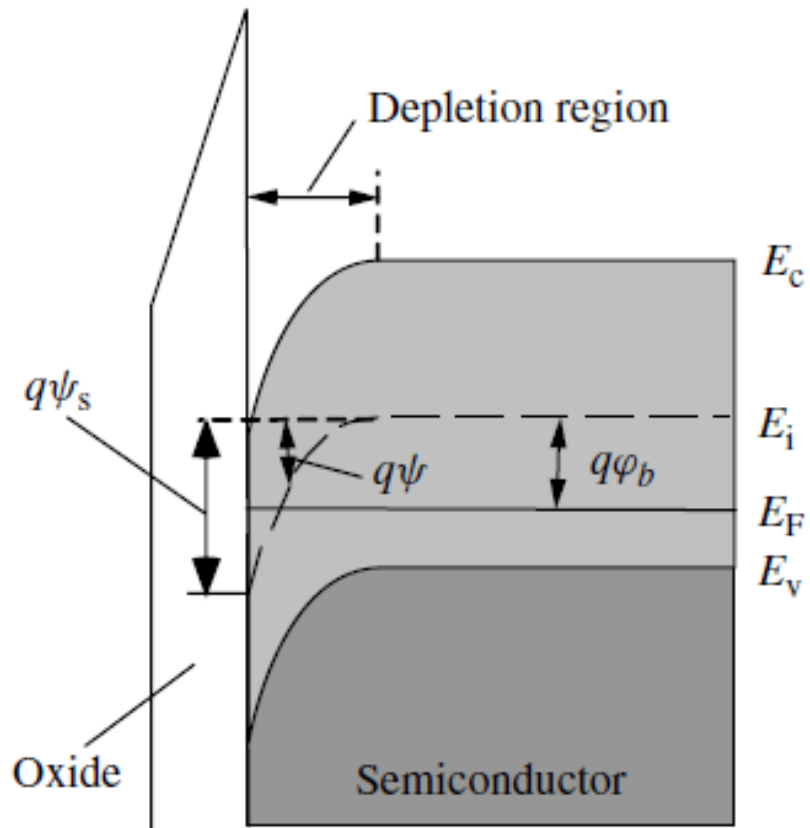


More holes accumulate at the surface of the semiconductor.



What happens if we keep increasing the amount of positive gate voltage we apply to the metal relative to the semiconductor?

Interface charge



When V_G is large enough, the surface is **inverted**.

The n-type surface that forms as a result of the applied electric field is the key to transistor operation!

Define a potential $q\psi_s$ which determines how much band bending there is at the surface.

- When $\psi_s = 0$ we are in flat band condition.
- When $\psi_s < 0$ we have hole accumulation at the surface.
- When $\psi_s > 0$ we have electron accumulation at the surface.
- When $\psi_s > \varphi_b$ we have inversion at the surface.
- **Surface should be as strongly n-type as the body is p-type.**

Figure 1.4 Band diagram for MOS capacitor in weak inversion ($\varphi_b < \psi_s < 2\varphi_b$)

$$\varphi_b = V_{th} \ln \left(\frac{N_a}{n_i} \right)$$

Strong inversion, when the total band bending equals $2q\varphi_b$, corresponding to the surface potential $\psi_s = 2\varphi_b$

Interface charge



The potential distribution $\psi(x)$ in the semiconductor can be determined from a solution of the one-dimensional Poisson's equation

$$\frac{d^2\psi(x)}{dx^2} = -\frac{\rho(x)}{\epsilon_s}$$

where ϵ_s is the semiconductor permittivity, and the space charge density $\rho(x)$ is given by

$$\rho(x) = q(p - n - N_a)$$

The position-dependent hole and electron concentrations may be expressed as follows using equilibrium statistics

$$p = N_a \exp(-\psi/V_{th}), \quad n_{po} = n_i^2/N_a \quad : \text{the equilibrium concentration of the minority carriers (electrons) in the bulk}$$
$$n = n_{po} \exp(\psi/V_{th}).$$

Note that deep inside the semiconductor, we have $\psi(\infty) = 0$

The surface concentrations of holes and electrons are expressed in terms of the surface potential as

$$p_s = N_a \exp(-\psi_s/V_{th}),$$
$$n_s = n_i^2/p_s = n_{po} \exp(\psi_s/V_{th})$$

Interface charge



In general, the above equations do not have an analytical solution for $\psi(x)$. However, the following expression can be derived for the electric field F_s at the insulator– semiconductor interface, in terms of the surface potential (see, e.g., Fjeldly *et al.*, Introduction to Device Modeling and Circuit Simulation 1998)

$$F_s = \sqrt{2} \frac{V_{th}}{L_{Dp}} f\left(\frac{\psi_s}{V_{th}}\right)$$

$$f(u) = \pm \sqrt{[\exp(-u) + u - 1] + \frac{n_{po}}{N_a} [\exp(u) - u - 1]},$$

and

$$L_{Dp} = \sqrt{\frac{\epsilon_s V_{th}}{q N_a}} \quad : \text{the Debye length}$$

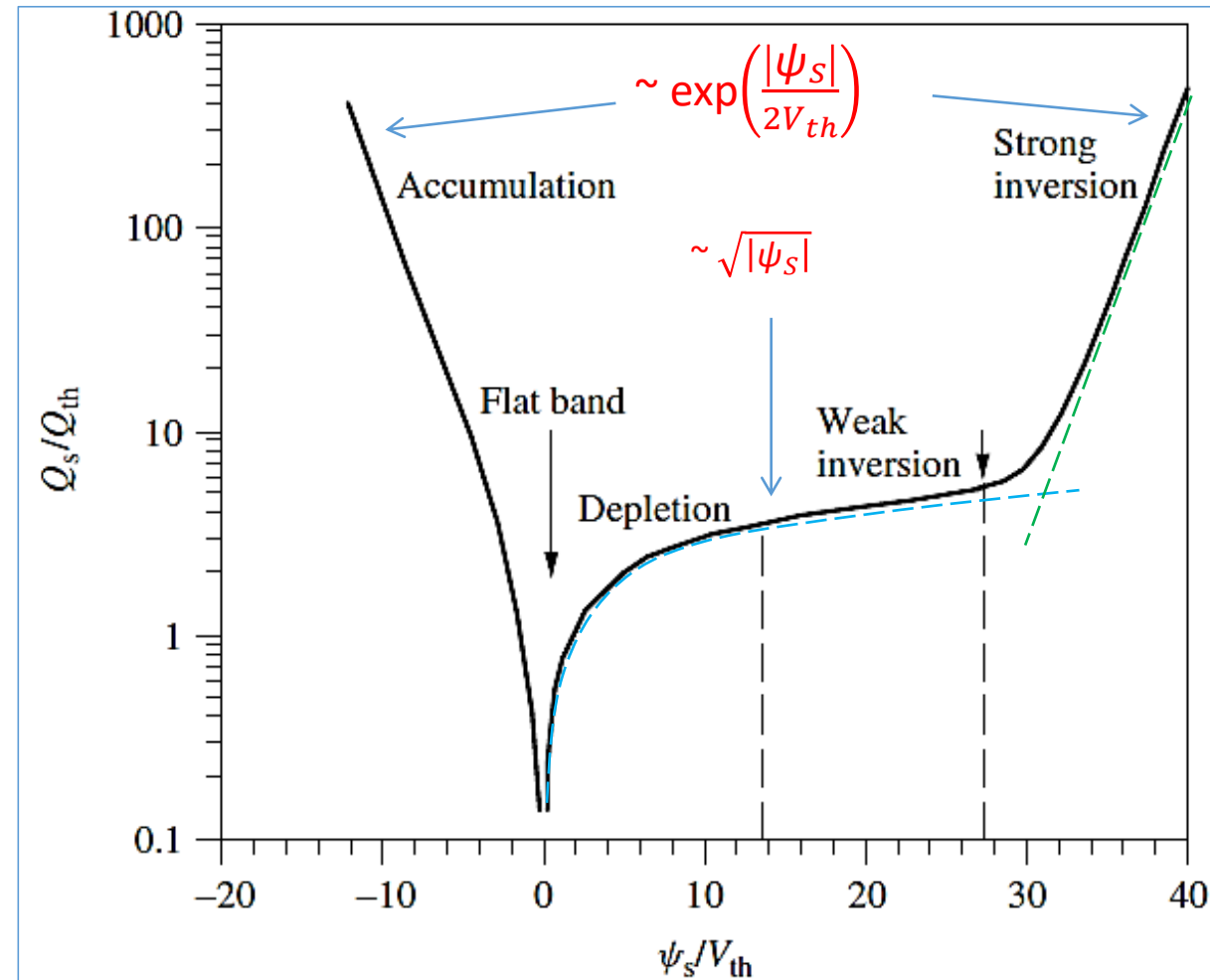
a positive sign should be chosen for a positive ψ_s and a negative sign corresponds to a negative ψ_s

Interface charge



Using Gauss' law, we can relate the total charge Q_s per unit area (carrier charge and depletion charge) in the semiconductor to the surface electric field by

$$Q_s = -\epsilon_s F_s.$$



Relation between Surface potential to applied voltage



In order to relate the semiconductor surface potential to the applied voltage V , we have to investigate how this voltage is divided between the insulator and the semiconductor. Using the condition of continuity of the electric flux density at the semiconductor–insulator interface, we find

$$\epsilon_s F_s = \epsilon_i F_i$$

where ϵ_i is the permittivity of the oxide layer and F_i is the constant electric field in the insulator (assuming no space charge). Hence, with an insulator thickness d_i , the voltage drop across the insulator becomes $F_i d_i$. Accounting for the flat-band voltage, the applied voltage can be written as

$$V = V_{FB} + \psi_s + \epsilon_s F_s / c_i$$

where $c_i = \epsilon_i / d_i$ is the insulator capacitance per unit area

Threshold Voltage



This charge increases gradually as the gate voltage is increased. However, this charge is only significant once the electron density at the surface exceeds the hole density in the substrate, N_a . We therefore define the threshold voltage as the gate voltage for which the electron density at the surface equals N_a . This corresponds to the situation where the total potential across the surface equals twice the bulk potential.

Strong inversion occurs when the surface potential ψ_s becomes equal to $2\phi_b$

The threshold voltage $V = V_T$, corresponding to the onset of the strong inversion, is one of the most important parameters characterizing metal-insulator-semiconductor devices.

We now derive the MOS parameters at threshold. To simplify the analysis we make the following assumptions

- 1) we assume that we can use the full depletion approximation, and
- 2) we assume that the inversion layer charge is zero below the threshold voltage.

Threshold Voltage



The derivation starts by examining the charge per unit area in the depletion layer, Q_d . This charge is given by:

$$Q_d = -qN_a x_d$$

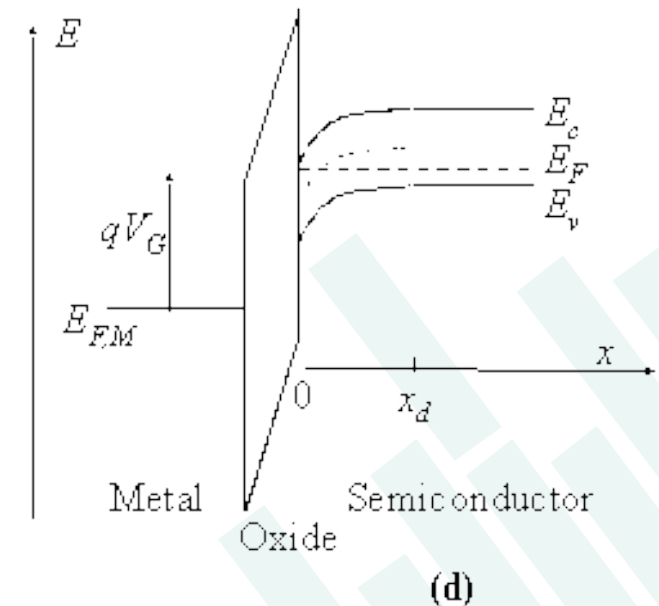
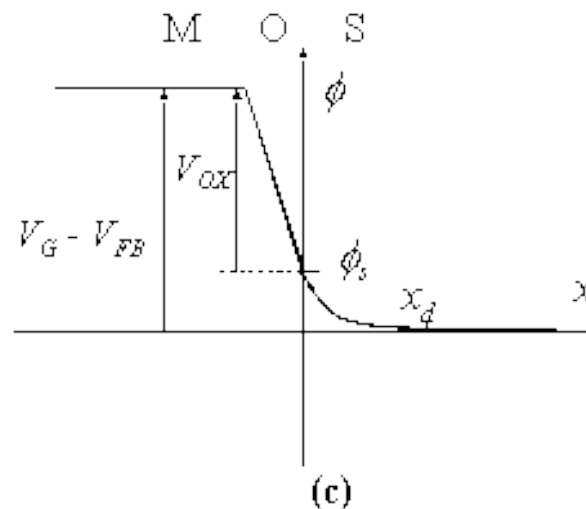
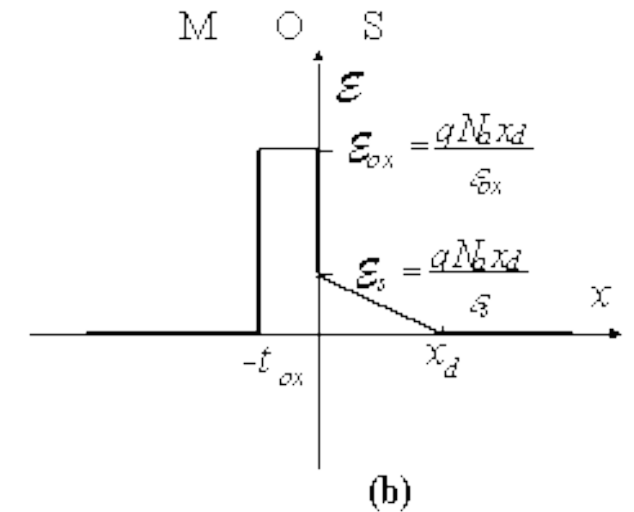
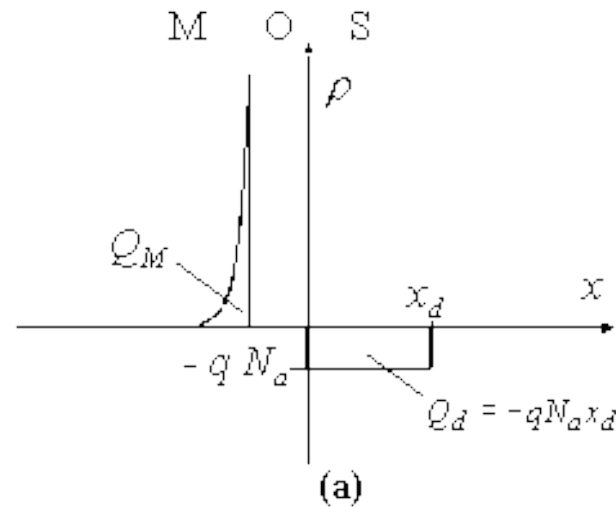
Where x_d is the depletion layer width and N_a is the acceptor density in the substrate.

And the Electric field at the semiconductor surface will be

$$F_s = \frac{qN_a x_d}{\epsilon_s}$$

The potential shown in Figure is obtained by integrating the electric field. The potential at the surface, ϕ_s , equals

$$\phi_s = \frac{qN_a x_d^2}{\epsilon_s}$$



Threshold Voltage

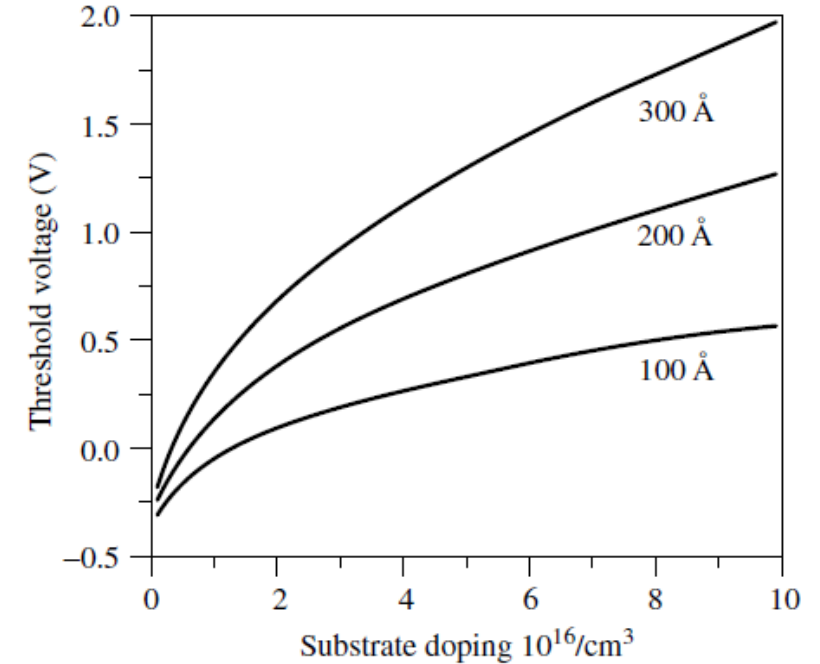


And the Electric field at the semiconductor surface for the threshold will be ($\psi_s = 2\phi_b$)

$$F_s^T = \sqrt{\frac{4qN_a\phi_b}{\epsilon_s}}$$

substituting the threshold values of ψ_s and F_s in the equation of voltage and surface potential relation (see three slides before)

$$V_T = V_{FB} + 2\phi_b + \sqrt{4\epsilon_s q N_a \phi_b / c_i}$$



Note: the threshold voltage may also be affected by so-called fast surface states at the semiconductor–oxide interface and by fixed charges in the insulator layer

The threshold voltage separates the subthreshold regime, where the mobile carrier charge increases exponentially with increasing applied voltage, from the above-threshold regime, where the mobile carrier charge is linearly dependent on the applied voltage.

MOS Capacitance



In a MOS capacitor, the metal contact and the neutral region in the doped semiconductor substrate are separated by the insulator layer, the channel, and the depletion region. Hence, the capacitance C_{mos} of the MOS structure can be represented as a series connection of the insulator capacitance $C_i = S(\epsilon_i/d_i)$, where S is the area of the MOS capacitor, and the capacitance of the active semiconductor layer C_s

$$C_{\text{mos}} = \frac{C_i C_s}{C_i + C_s}$$

The semiconductor capacitance can be calculated as

$$C_s = S \left| \frac{dQ_s}{d\psi_s} \right| \quad \text{where } Q_s \text{ is the total charge density per unit area in the semiconductor and } \psi_s \text{ is the surface potential}$$

Using Q_s from previous calculations and performing the differentiation, we obtain

$$C_s = \frac{C_{so}}{\sqrt{2} f(\psi_s/V_{th})} \left\{ 1 - \exp\left(-\frac{\psi_s}{V_{th}}\right) + \frac{n_{po}}{N_a} \left[\exp\left(\frac{\psi_s}{V_{th}}\right) - 1 \right] \right\}$$

Here, $C_{so} = S(\epsilon_s/L_{Dp})$ is the semiconductor capacitance at the flat-band condition (i.e., for $\psi_s = 0$) and L_{Dp} is the Debye length

MOS Capacitance



The semiconductor capacitance can formally be represented as the sum of two capacitances – a depletion layer capacitance C_d and a free carrier capacitance C_{fc}

The depletion layer capacitance is given by $C_d = S\epsilon_s/d_d$ where $d_d = \sqrt{\frac{2\epsilon_s\psi_s}{qN_a}}$ is the depletion layer width

The free carrier contribution to the semiconductor capacitance can be formally expressed as

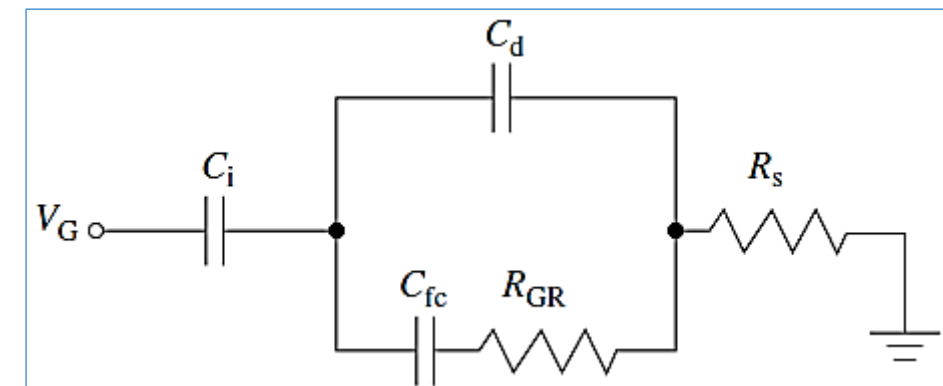
$$C_{fc} = C_s - C_d$$

The capacitance C_{fc} becomes important in the inversion regime, especially in strong inversion where the mobile charge is important

The variation in minority carrier charge at the semiconductor–insulator interface therefore proceeds at a rate limited by the time constants associated with the generation/recombination processes

This finite rate represents a delay, which may be represented electrically in terms of an RC product consisting of the capacitance C_{fc} and the resistance R_{GR}

This equivalent circuit is clearly frequency-dependent



MOS Capacitance



In the low-frequency limit, we can neglect the effects of R_{GR} and R_s to obtain

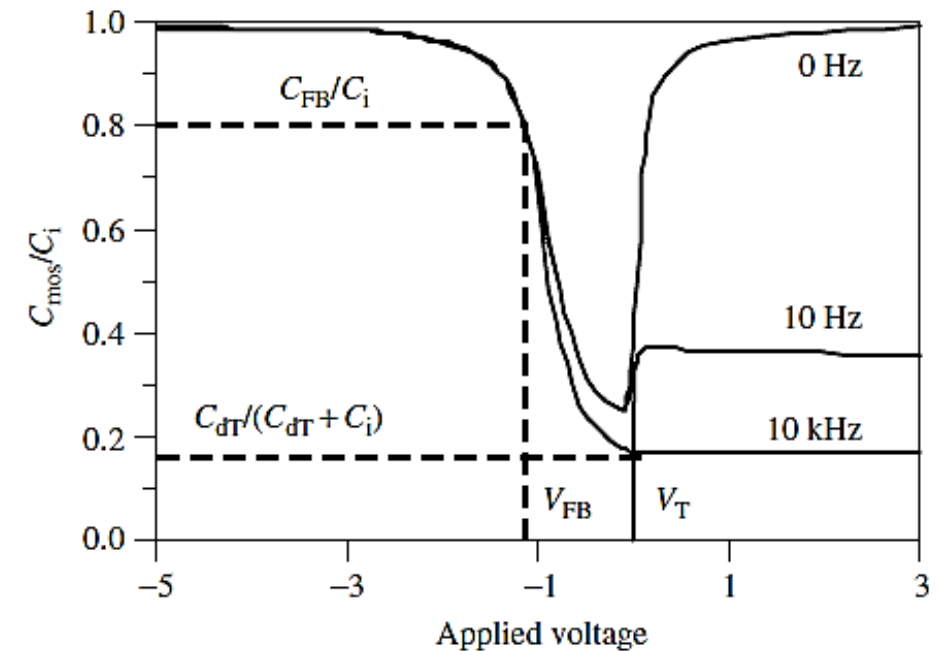
$$C_{mos}^0 = \frac{C_s C_i}{C_s + C_i}$$

In strong inversion, we have $C_s \gg C_i$, which gives

$$C_{mos}^0 \approx C_i$$

In the high-frequency limit, the time constant of the generation/recombination mechanism will be much longer than the signal period ($R_{GR} C_{fc} \gg 1/f$) and C_d effectively shunts the lower branch of the parallel section of the equivalent

$$C_{mos}^\infty = \frac{C_{dT} C_i}{C_{dT} + C_i}$$



From MOS C-V characteristics, we can easily determine the gate insulator thickness, the semiconductor substrate doping density, and the flat-band voltage, etc.

MOS Capacitance



The maximum measured capacitance $C_{\max}$ (capacitance C_i in previous Figure) yields the insulator thickness

$$d_i \approx S\epsilon_i / C_{\max}$$

The minimum measured capacitance $C_{\min}$ (at high frequency) allows us to find the doping concentration in the semiconductor substrate. First, we determine the depletion capacitance in the strong inversion regime using

$$1/C_{\min} = 1/C_{dT} + 1/C_i$$

From C_{dT} we obtain the thickness of the depletion region at threshold as $d_{dT} = S\epsilon_s / C_{dT}$

Then we calculate the doping density N_a with $\psi_s = 2\phi_b$. This results in the following transcendental equation for N_a

$$N_a = \frac{4\epsilon_s V_{th}}{q d_{dT}^2} \ln \left(\frac{N_a}{n_i} \right)$$

This equation can easily be solved by iteration or by approximate analytical techniques

Once d_i and N_a have been obtained, the device capacitance CFB under flat-band conditions can be determined using $C_s = C_{so}$ (at flat-band condition),

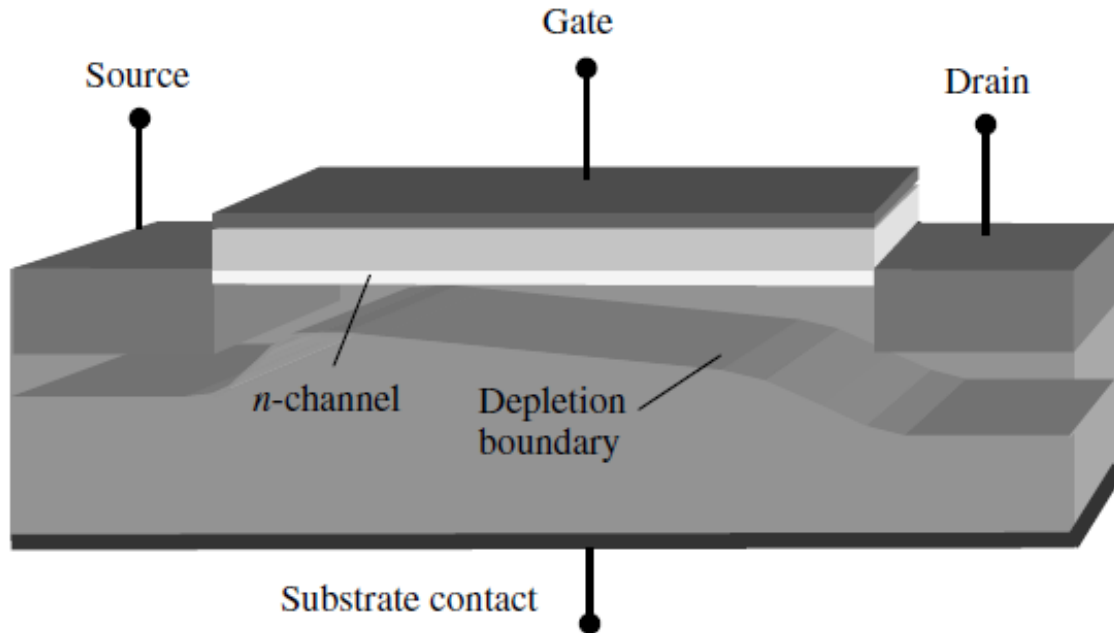
$$C_{FB} = \frac{C_{so} C_i}{C_{so} + C_i} = \frac{S\epsilon_s \epsilon_i}{\epsilon_s d_i + \epsilon_i L_{Dp}}$$

The above characterization technique applies to ideal MOS structures. Different nonideal effects, such as geometrical effects, nonuniform doping in the substrate, interface states, and mobile charges in the oxide may influence the C-V characteristics of the MOS capacitor

MOSFET



An inversion layer at the semiconductor–oxide interface acts as a conducting channel. For example, in an n -channel MOSFET, the substrate is p -type silicon and the inversion charge consists of electrons.



When a drain-source bias V_{DS} is applied to an n -channel MOSFET in the above threshold conducting state, electrons move in the channel inversion layer from source to drain.

Requirement to assure that the induced inversion channel extends all the way from source to drain:

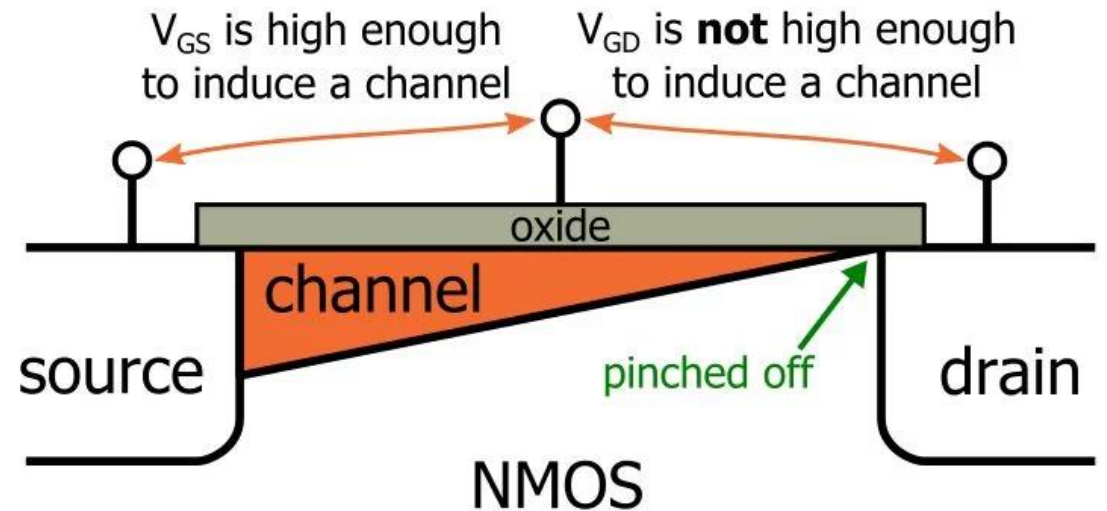
The MOSFET gate structure either overlaps slightly or aligns with the edges of these contacts (achieved by a self-aligned process). Self-alignment is preferable since it minimizes the parasitic gate-source and gate-drain capacitances.

MOSFET



For $V_{GS} > V_T$ in an n-channel device, the application of a positive V_{DS} gives a steady voltage increase from source to drain along the channel that causes a corresponding reduction in the local gate-channel bias V_{GX} (here X signifies a position x within the channel). This reduction is greatest near drain where V_{GX} equals the gate-drain bias V_{GD}

We may say that when $V_{GD} = V_T$, the channel reaches threshold at the drain and the density of inversion charge vanishes at this point. This is the so-called **pinch-off condition**, which leads to a saturation of the drain current I_{ds} .



The corresponding drain-source voltage, $V_{DS} = V_{SAT}$, is called the saturation voltage

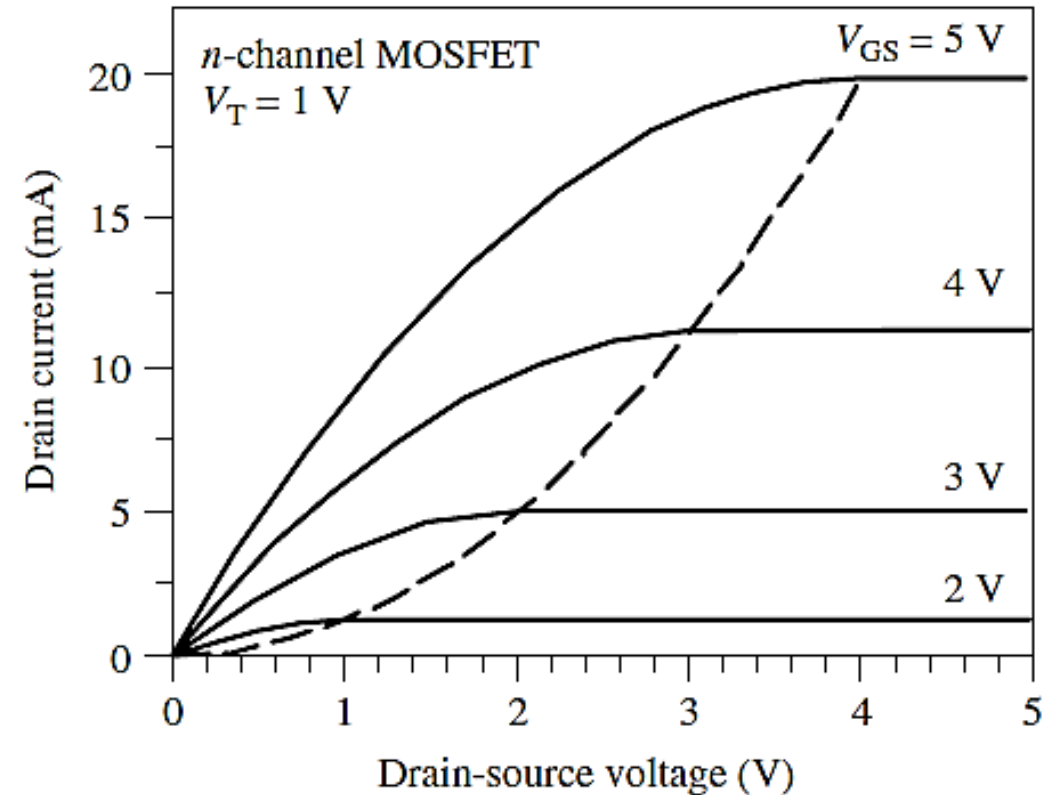
Since $V_{GD} = V_{GS} - V_{DS}$, we find that $V_{SAT} = V_{GS} - V_T$

MOSFET



When $V_{DS} > V_{SAT}$, the pinched-off region near drain expands only slightly in the direction of the source, leaving the remaining inversion channel intact. Hence, the drain current in saturation remains approximately constant, given by the voltage drop V_{SAT} across the part of the channel that remain in inversion. The voltage ($V_{DS} - V_{SAT}$) across the pinched-off region creates a strong electric field, which efficiently transports the electrons from the strongly inverted region to the drain.

Typical current–voltage characteristics of a long-channel MOSFET, where pinch-off is the predominant saturation mechanism, are shown in this Figure



Basic MOSFET modeling



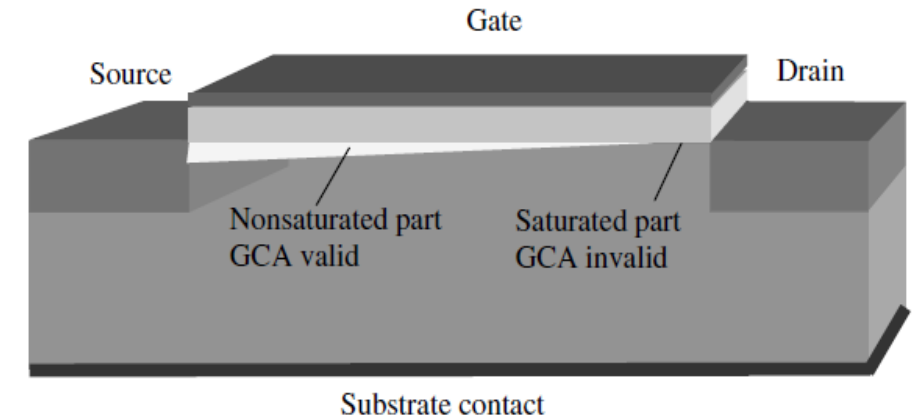
Charge Control Model (valid for long-channel MOSFETs)

Based on gradual channel approximation (CGA) (when $E_y \gg E_x$)

Above the threshold voltage, the absolute value of the drain current can be written as:

$$I_{ds} = W \mu_n q n_s F$$

where $F = |dV/dx|$ is the magnitude of the electric field in the channel and W is the channel width.



Integrating this expression over the gate length and using the fact that I_{ds} is independent of position x , we obtain the following expression for the current–voltage characteristics:

$$I_{ds} = \frac{W \mu_n C_i}{L} \times \begin{cases} (V_{GT} - V_{DS}/2) V_{DS}, & \text{for } V_{DS} \leq V_{SAT} = V_{GT} \\ V_{GT}^2/2, & \text{for } V_{DS} > V_{SAT} \end{cases}$$

where $V_{GT} \equiv V_{GS} - V_T$

Basic MOSFET modeling



Important device parameters are the channel conductance

$$g_d = \left. \frac{\partial I_d}{\partial V_{DS}} \right|_{V_{GS}} = \begin{cases} \beta(V_{GT} - V_{DS}), & \text{for } V_{DS} \leq V_{SAT} \\ 0, & \text{for } V_{DS} > V_{SAT} \end{cases}$$

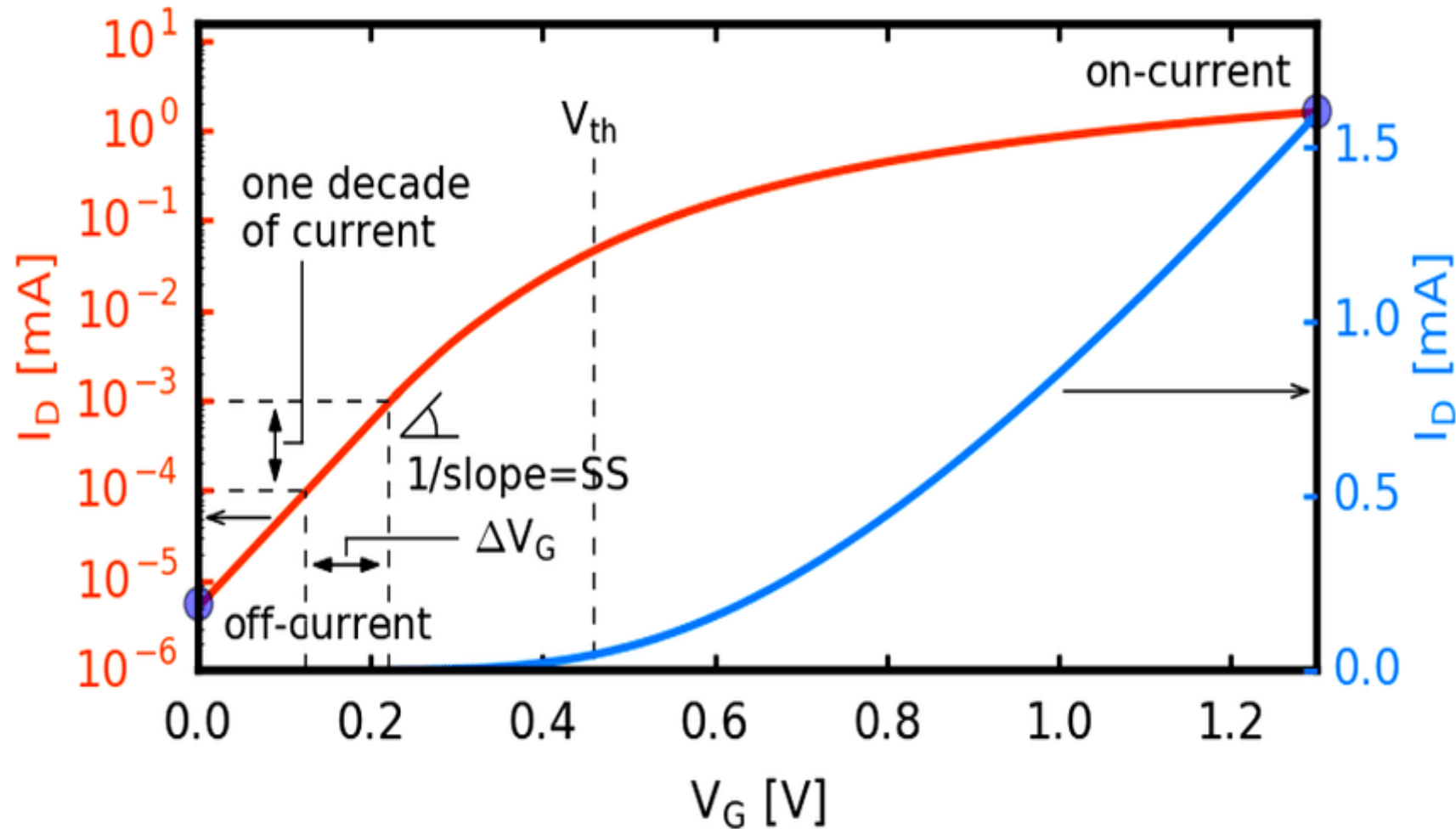
and the transconductance

$$g_m = \left. \frac{\partial I_d}{\partial V_{GS}} \right|_{V_{DS}} = \begin{cases} \beta V_{DS}, & \text{for } V_{DS} \leq V_{SAT} \\ \beta V_{GT}, & \text{for } V_{DS} > V_{SAT} \end{cases}$$

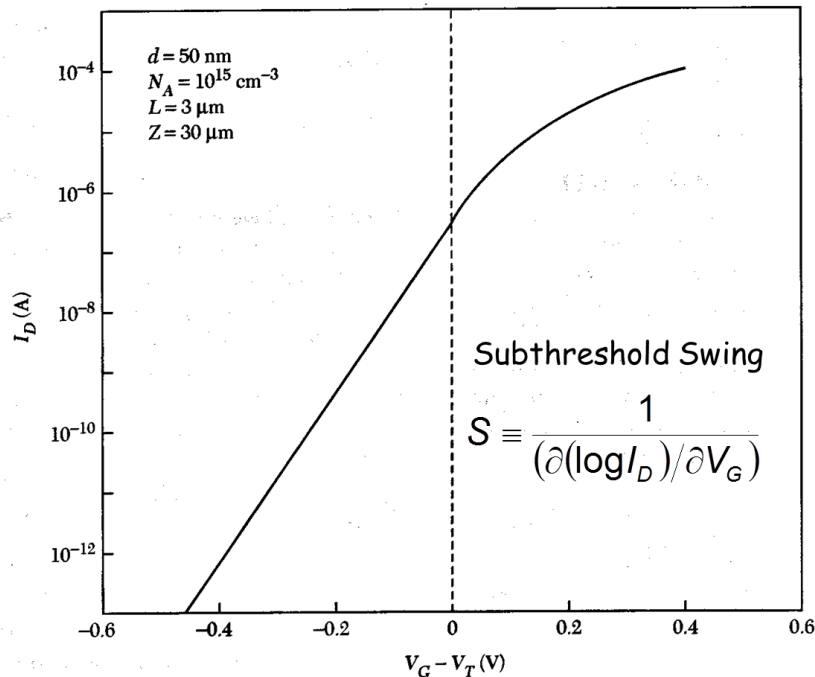
where $\beta = W\mu_n c_i/L$ is called the transconductance parameter

As can be seen from these expressions, high values of channel conductance and transconductance are obtained for large electron mobilities, large gate insulator capacitances (i.e., thin gate insulator layers), and large gate width to length ratios.

MOSFET transfer characteristics



MOSFET transfer characteristics



For gate voltage less than the threshold – weak inversion → **Diffusion is dominant current mechanism (not drift)**

$$I_D = J_D A = -qAD_n \frac{\partial n}{\partial y} \approx -qAD_n \frac{n(0) - n(L)}{L}$$

$$n(0) = n_i e^{q(\psi_s - \psi_B)/kT}$$

$$n(L) = n_i e^{q(\psi_s - \psi_B - V_D)/kT}$$

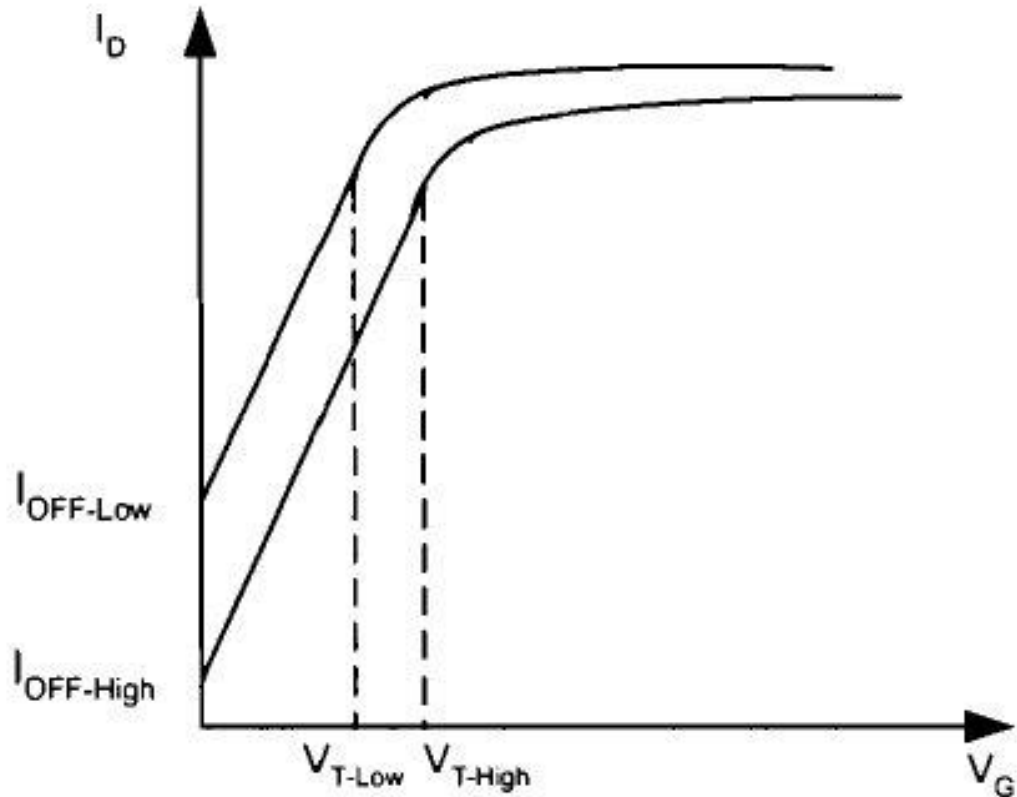
$$I_D = \frac{qAD_n n_i e^{-\psi_B/kT}}{L} \left(1 - e^{-qV_D/kT}\right) e^{q\psi_s/kT}$$

$$I_D \approx \frac{qAD_n n_i e^{-\psi_B/kT}}{L} \left(1 - e^{-qV_D/kT}\right) e^{q(V_G - V_T)/kT}$$

We can approximate ψ_s with $(V_G - V_T)$ below threshold since all voltage drops across depletion region

- Sub-threshold current is exponential function of applied gate voltage
- Sub-threshold current gets larger for smaller gates (L)

impact of reduced threshold voltage on leakage current



Now since I_{DS} is proportional to $(V_{GS} - V_T)$, therefore as V_T begins to fall, I_{DS} starts increasing resulting in larger drain currents. However *due to the restriction of SS to 60mV/dec, I_{OFF} increases rapidly.*

MOSFETs

- Enhancement mode

- Also known as **Normally Off transistors**.
 - A voltage must be applied to the gate of the transistor, at least equal to the threshold voltage, to create a conduction path between the source and the drain of the transistor before current can flow between the source and drain.

- Depletion mode

- Also known as **Normally On transistors**.
 - A voltage must be applied to the gate of the transistor, at least equal to the threshold voltage, to destroy a conduction path between the source and the drain of the transistor to prevent current from flowing between the source and drain.

